PS3:EXOPLANET HUNTERS

SUBMITTED BY TEAM:CLOCK IT

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A: DETECT THE EXOPLANETS

$\delta = (F_{normal} - F_{transit}) / F_{normal}$

	Transiting exoplanet	DIPS	Orbital period (Days)	TRANSIT DEPTH
STAC 1	YES	3,8,12,16,20	4	0.003
STAC 2	YES	20,100,180	80	0.001
STAC 3	NOT SURE	20,60,100,140,170	NOT PERIODIC	0.001, 0.0009, 0.0008, 0.0007, 0.001

4-A periodic decrease in a star's brightness can indicate the presence of an exoplanet because when a planet passes in front of its host star, it blocks a small portion of the star's light. This event is called a transit. If the brightness decreases at regular intervals, it suggests that the planet is orbiting the star and repeatedly passing in front of it. The time between consecutive transits can be used to estimate the planet's orbital period.

STAC 1*:interval between consecutive dips –*

8-3=5 days
12-8=4days
16-12=4days
20-16=4days

The first interval is 5 days but the remaining intervals are consistently 4 days**. Hence, the approx. orbital period is 4 days.**

STAC 2:the information given has big intervals so we cannot conclude if the orbital period is 80 days or not

Possible orbital periods= 10,20,40,80

For 10 days-

We would expect the transit to be 20,30,40…..

The 40,60 etc do not have a brightness dip so it can't be the orbital period same reasoning with 20 and 40

Therefore the orbital period is 80 days

STAC3:the intervals between dips

60-20=40 days
100-60=40 days
140-100=40 days
170-140=30 days

This does not establish a consistent orbital period.

Ps: this does not prove that stac 3 has no planet. It means the available observations are insufficient to confirm a periodically transiting planet.

B: DETERMINE THE PLANET SIZE

$\delta \approx (Rp / Rs)*2$

Information provided:
Radius of super earth=6371km-12742km
Radius of Neptune=24764km
Radius of Jupiter = 69911km
Solar radius($R_\odot$)=696,000km

STAC 1:

$(\delta)=0.003$
$R_P = R_S\sqrt{\delta}$
$R_p=696{,}000\text{km}\sqrt{0.003}$
$R_P=38{,}121.49\text{km}$
$R_P=5.98R_E$
Hence, it is approximately Neptune sized.

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STAC 2:

$(\delta)=0.001$
$R_P = R_S\sqrt{\delta}$
$R_S=0.8\ R_\odot$
$R_p=696{,}000\times 0.8\times\sqrt{0.001}$
$R_P=17{,}605\text{km}$
$Rp=2.76Re$

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Hence, this is approximately super earth sized.

STAC 3:

$(\delta 1)=0.001$

$(\delta 2)=0.0008$

$(\delta 3)=0.0009$

$(\delta 4)=0.0007..$

$(\delta 5)=0.001$

$(\delta avg)=0.00088$

$R_P = R_S\sqrt{\delta}$

$R_S = 0.5\ R_\odot$

$R_P = 696{,}000 \times 0.5 \times \sqrt{0.00088}$

$R_P = 10{,}322 km$

$R_P = 1.62 R_E$

It is also approximately super earth sized

C: DETERMINE THE PLANETARY ORBITS

P^2 = a^3 / M

P = orbital period in years

a = orbital distance in AU M = mass of the star in solar masses

STAC 1:

P=4 DAYS=4/365.25years=0.010951 years

P^2=a^3/M

Calculate orbital distance(a)

$$a=\sqrt[3]{p^2 \times M}$$

substituting values we get

a=0.0493AU

1AU=149.6 million km

a=7.73million km

earths orbital distance is 1AU

for stac1 the orbital distance is 0.0493

therefore the the planets orbit is much closer to its star than earths orbits the sun

STAC 2:

P=80 days=80/365.25=0.2190years

$$a=\sqrt[3]{p^2 \times M}$$

substituting values we get

a=0.3365AU

a=50.34 million km

stac2 planet orbits at approximately 0.3360AU therefore 33.65% of earth's orbital distance.

STAC 3:

Assume that P=40 because stac3 has dips repeating approximately every 40 days but day 170 dip is inconsistent.

P=40/365.25=0.1095

$$a=\sqrt[3]{p^2 \times M}$$

substituting values we get,

a=0.182AU

a=0.182X149.6million km

a=27.2 million km

comparing with earth's distance from the sun,stac 3 planet would orbit at approximately,18.2%of earth's distance from the sun

D. FIND THE MOST PROMISING WORLD

$d_{HZ} \approx \sqrt{(L / L\odot)}\ AU$

d_{HZ} = approximate distance of the habitable zone from the star

L = luminosity of the star

$L\odot$ = luminosity of the Sun

STAC1:

$L=1.00\ L\odot$

$D_{hz}=\sqrt{1.00}=1AU$

STAC2:

$L=0.40\ L\odot$

$D_{hz}=\sqrt{0.40}=0.6321AU$

STAC3:

$L=0.08\ L\odot$

$D_{hz}=\sqrt{0.08}=0.283AU$

SYSTEM	Planet's orbital distance	Approx. habitable zone distance
STAC 1	0.049AU	1AU
STAC 2	0.337AU	0.632AU
STAC 3	0.182AU	0.283AU

STAC 1

0.049AU<1AU

Stac1 planet is much closer to its star than the approx. habitatble zone

It is unlikely to be most promising candidate for surface habitability because it orbits close to the surface of the earth

STAC 2

0.392AU<0.632AU

Stac2 's planet orbits inside the approx habitable zone distance calculated for its star. It is therefore not as promising as a planet orbiting near that distance

STAC 3

Stac 3 does not show a sufficiently consistent periodic transit signal in the provided observations.Therefore,we cannot reliably calculate an orbital period or orbital distance for its suspected planet.

FINAL CONCLUSION:

Stac 2 is the most promising candidate for further habitability studies among the detected planets.

THE EVIDENCE

[1] **Longer orbital period:** STAC-2 has an orbital period of approximately 80 days, compared with approximately 4 days for STAC-1. This suggests that STAC-2's planet is farther from its star.

[2] **Greater orbital distance:** STAC-2's orbital distance is approximately 0.338 AU, whereas STAC-1's is approximately 0.049 AU. STAC-2's planet is therefore much farther from its host star.

[3] **Cooler host star:** STAC-2's star has a surface temperature of 5000 K, compared with 5800 K for STAC-1. Its lower luminosity also means that its habitable zone is closer to the star.

[4] **Closer to its habitable zone relative to the other detected planet:** STAC-2 orbits at 0.338 AU, compared with its estimated habitable-zone distance of 0.632 AU. STAC-1 orbits at only 0.049 AU, far inside its estimated habitable-zone distance of 1 AU

Limitation: STAC-2 is still inside the approximate inner boundary implied by the simplified habitable-zone relation. Therefore, we cannot conclude that it is habitable. It is simply the better candidate for further investigation based on the available data.

ADDITIONAL OBSERVATION

An additional observation would be to study the star's light as it passes through the planet's atmosphere during a transit. Different gases absorb different wavelengths of light, which can help identify the gases present in the atmosphere. This would help us determine whether the planet has an atmosphere and whether it contains gases that could be suitable for life.

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